

LAB 4 – SQL FUNCTIONS:

Conversion Functions: -

```
> CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    email VARCHAR(100) UNIQUE,  
    hire_date DATE,  
    department_id INT  
);
```

```
INSERT INTO employees VALUES
```

```
(101, 'Alice', 'Johnson', 'alice.j@company.com', '2020-05-10', 1),  
(102, 'Bob', 'Smith', 'bob.s@company.com', '2019-03-15', 2),  
(103, 'Charlie', 'Brown', 'charlie.b@company.com', '2021-07-20', 3),  
(104, 'Diana', 'Lee', 'diana.l@company.com', '2018-11-01', 4),  
(105, 'Ethan', 'Miller', 'ethan.m@company.com', '2022-01-12', 2),  
(106, 'Fiona', 'Davis', 'fiona.d@company.com', '2020-09-23', 3),  
  
(107, 'George', 'Wilson', 'george.w@company.com', '2021-04-18', 1),  
(108, 'Hannah', 'Moore', 'hannah.m@company.com', '2019-08-30', 4),  
(109, 'Ivan', 'Taylor', 'ivan.t@company.com', '2022-11-05', 3),  
(110, 'Julia', 'Anderson', 'julia.a@company.com', '2023-02-10', 2);
```

```
SELECT * FROM employees;
```

employee_id	first_name	last_name	email	hire_date	department_id
101	Alice	Johnson	alice.j@company.com	2020-05-10	1
102	Bob	Smith	bob.s@company.com	2019-03-15	2
103	Charlie	Brown	charlie.b@company.com	2021-07-20	3
104	Diana	Lee	diana.l@company.com	2018-11-01	4
105	Ethan	Miller	ethan.m@company.com	2022-01-12	2
106	Fiona	Davis	fiona.d@company.com	2020-09-23	3
107	George	Wilson	george.w@company.com	2021-04-18	1
108	Hannah	Moore	hannah.m@company.com	2019-08-30	4
109	Ivan	Taylor	ivan.t@company.com	2022-11-05	3
110	Julia	Anderson	julia.a@company.com	2023-02-10	2
NULL	NULL	NULL	NULL	NULL	NULL

Implicit Data Type Conversion:

```
SELECT employee_id, first_name, department_id
FROM employees
WHERE department_id > 2;
```

	employee_id	first_name	department_id
▶	103	Charlie	3
	104	Diana	4
	106	Fiona	3
	108	Hannah	4
	109	Ivan	3
*	NULL	NULL	NULL

Explicit Data Type Conversion:

```
SELECT employee_id, first_name,
       CAST(hire_date AS CHAR) AS hire_date_text
FROM employees;
```

employee_id	first_name	hire_date_text
101	Alice	2020-05-10
102	Bob	2019-03-15
103	Charlie	2021-07-20
104	Diana	2018-11-01
105	Ethan	2022-01-12
106	Fiona	2020-09-23
107	George	2021-04-18
108	Hannah	2019-08-30
109	Ivan	2022-11-05
110	Julia	2023-02-10

Date Functions: -

```

SELECT
    first_name,
    last_name,
    hire_date,
    CURDATE() AS today,
    YEAR(hire_date) AS year,
    MONTH(hire_date) AS month,
    DAY(hire_date) AS day,
    YEAR(CURDATE()) - YEAR(hire_date) AS approximate_years
FROM employees;

```

first_name	last_name	hire_date	today	year	month	day	approximate_years
Alice	Johnson	2020-05-10	2026-10-02	2020	5	10	6
Bob	Smith	2019-03-15	2026-10-02	2019	3	15	7
Charlie	Brown	2021-07-20	2026-10-02	2021	7	20	5
Diana	Lee	2018-11-01	2026-10-02	2018	11	1	8
Ethan	Miller	2022-01-12	2026-10-02	2022	1	12	4
Fiona	Davis	2020-09-23	2026-10-02	2020	9	23	6
George	Wilson	2021-04-18	2026-10-02	2021	4	18	5
Hannah	Moore	2019-08-30	2026-10-02	2019	8	30	7
Ivan	Taylor	2022-11-05	2026-10-02	2022	11	5	4
Julia	Anderson	2023-02-10	2026-10-02	2023	2	10	3

LAB 5 – SQL OPERATORS: -

```

SELECT
    employee_id,
    first_name,
    department_id,

    employee_id + 10 AS id_plus_10,

    employee_id > 105 AS id_greater_than_105,

    department_id = 2 AND employee_id > 105 AS dept2_and_id_above_105,
    department_id = 2 OR department_id = 3 AS dept2_or_dept3,
    NOT department_id = 2 AS not_dept2

FROM employees;

```

employee_id	first_name	department_id	id_plus_10	id_greater_than_105	dept2_and_id_above_105	dept2_or_dept3	not_dept2
101	Alice	1	111	0	0	0	1
102	Bob	2	112	0	0	1	0
103	Charlie	3	113	0	0	1	1
104	Diana	4	114	0	0	0	1
105	Ethan	2	115	0	0	1	0
106	Fiona	3	116	1	0	1	1
107	George	1	117	1	0	0	1
108	Hannah	4	118	1	0	0	1
109	Ivan	3	119	1	0	1	1
110	Julia	2	120	1	1	1	0

LAB 6 – SPECIAL OPERATORS & SET OPERATIONS: -

Special Operators:

```

SELECT
    employee_id,
    first_name,
    department_id,
    email
FROM employees
WHERE employee_id BETWEEN 103 AND 108
    OR department_id IN (1, 3)
    OR first_name LIKE 'A%'
    OR department_id IS NULL;

```

employee_id	first_name	department_id	email
101	Alice	1	alice.j@company.com
103	Charlie	3	charlie.b@company.com
104	Diana	4	diana.l@company.com
105	Ethan	2	ethan.m@company.com
106	Fiona	3	fiona.d@company.com
107	George	1	george.w@company.com
108	Hannah	4	hannah.m@company.com
109	Ivan	3	ivan.t@company.com
NULL	NULL	NULL	NULL

Set Operations:

Union –

```
SELECT first_name  
FROM employees  
WHERE department_id = 1
```

UNION

```
SELECT first_name  
FROM employees  
WHERE department_id = 3;
```

	first_name
▶	Alice
	George
	Charlie
	Fiona
	Ivan

Intersect –

```
SELECT department_id  
FROM employees  
WHERE employee_id <= 105
```

INTERSECT

```
SELECT department_id  
FROM employees  
WHERE employee_id >= 103;
```

	department_id
▶	1
	2
	3
	4